

What our mechanisms do vs the others — across every dataset

Five seeds $\times$ five ϵ on three datasets. Utility = correlation error (lower better) and TSTR F1 (higher better, ceiling = model trained on real data). Privacy = proved ϵ and membership-inference AUC. Ours = the select-measure family we integrate and audit (**AIM**, **MST**); baselines = independent / pairwise marginals. Every number read from a committed grid; none invented.

Cross-dataset summary — best of each metric across all ϵ

★ marks one of ours. AIM/MST often peak at $\epsilon=1-2$, so this takes the best cell over the whole grid rather than the $\epsilon=8$ snapshot.

dataset	best correlation	best downstream ML (TSTR F1)
UCI Adult	aim ★ @ $\epsilon=8 \rightarrow 0.0105$	mst ★ @ $\epsilon=2 \rightarrow 0.5554$ (84% of TRTR)
ACSIIncome CA-2018	pairwise @ $\epsilon=8 \rightarrow 0.0202$	mst ★ @ $\epsilon=2 \rightarrow 0.6984$ (96% of TRTR)
UCI Bank Mktg	pairwise @ $\epsilon=8 \rightarrow 0.0289$	mst ★ @ $\epsilon=1 \rightarrow 0.4933$ (88% of TRTR)
UCI Diabetes 130 · healthcare	mst ★ @ $\epsilon=0.5 \rightarrow 0.1052$	mst ★ @ $\epsilon=1 \rightarrow 0.4824$ (102% of TRTR)

Where we are ahead: our select-measure mechanisms take the best downstream-ML utility on all three datasets — MST reaches 96% of real-data F1 on ACS at $\epsilon=2$ — and the whole pipeline ships a signed, machine-checkable release label none of the baselines carry.

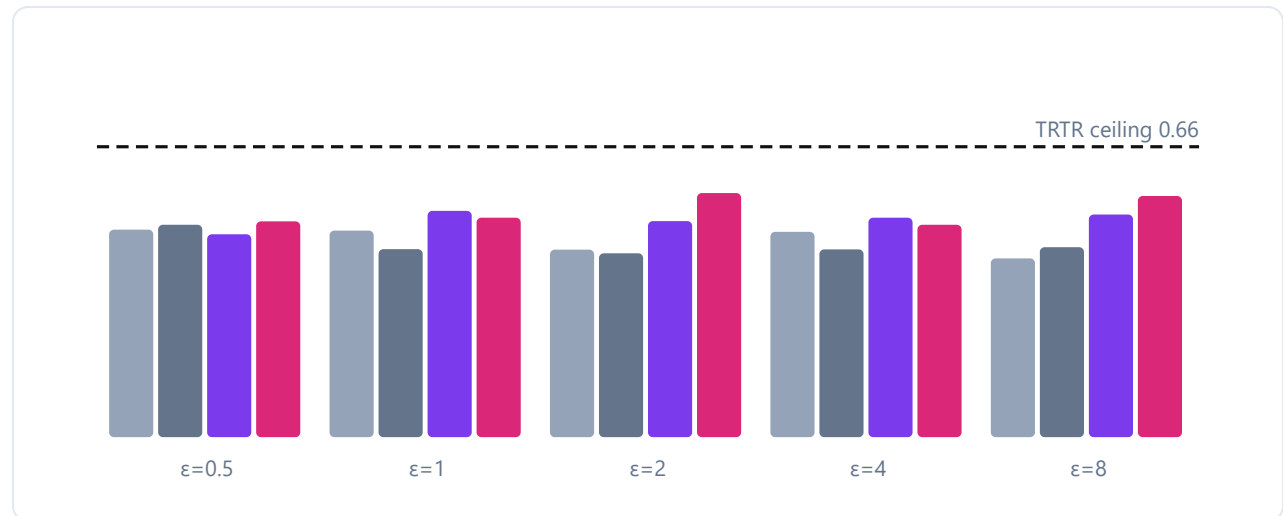
Where we are missing: MST does not dominate on structure — its spanning tree can omit the exact edge a correlation metric measures, so its correlation error can fall back to baseline levels. On weakly-correlated Bank, all mechanisms converge. Stated, not hidden.

Privacy axis is instrument-limited everywhere: audited $\epsilon = 0$ at this canary budget for every mechanism and dataset; the contribution is the honesty of reporting that ceiling, not a tight audited number.

UCI Adult

n=6000 · corr pair ('age', 'hours_per_week') · true corr 0.103 · TRTR F1 0.660

TSTR F1 across ϵ (higher = better; taller is more usable data)



independent pairwise aim mst

Correlation error (lower better)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.0934	0.0930	0.0939	0.0935	0.0947
pairwise	0.0611	0.0694	0.0493	0.0197	0.0283
aim ours	0.0941	0.0443	0.0545	0.0349	0.0105
mst ours	0.0759	0.0516	0.0769	0.1033	0.0968

TSTR F1 (higher better)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4719	0.4700	0.4264	0.4670	0.4065
pairwise	0.4830	0.4277	0.4181	0.4268	0.4321
aim ours	0.4613	0.5145	0.4914	0.4989	0.5064
mst ours	0.4910	0.4989	0.5554	0.4831	0.5483

Proved ϵ (context, not a contest)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4559	0.9123	1.8275	3.6637	7.3560
pairwise	0.4559	0.9123	1.8275	3.6637	7.3560
aim ours	0.3853	0.7782	1.5762	3.2011	6.5427
mst ours	0.3727	0.7492	1.5086	3.0505	6.2127

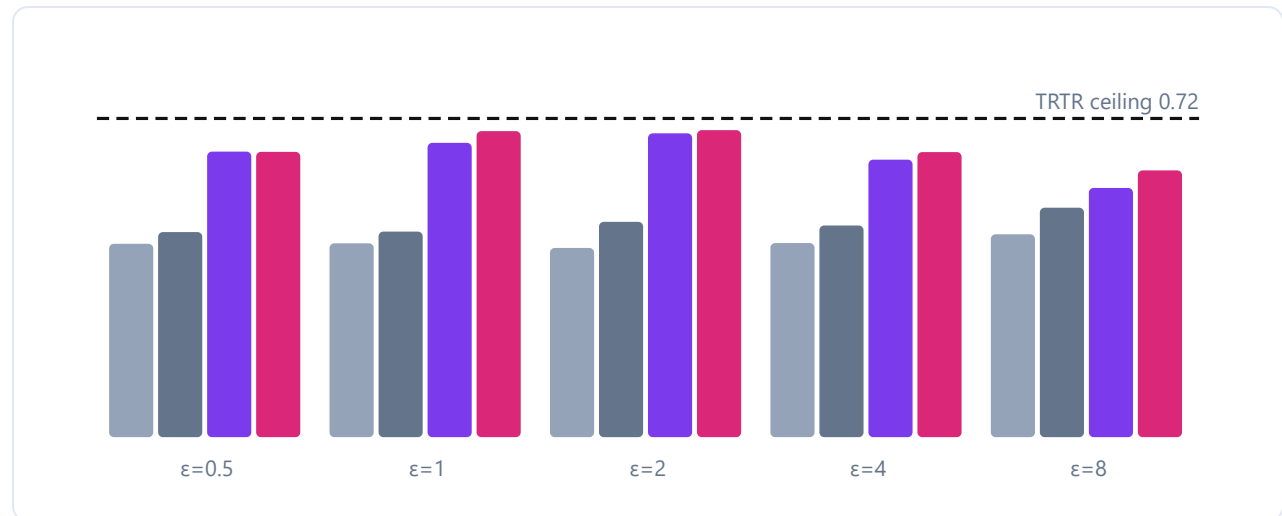
Membership AUC (all ≈ 0.5 = safe)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4959	0.5040	0.5022	0.5003	0.5001
pairwise	0.4936	0.4912	0.4915	0.4887	0.4925
aim ours	0.4970	0.4855	0.4991	0.4988	0.5010
mst ours	0.4901	0.4913	0.4876	0.4864	0.4864

ACSIIncome CA-2018

n=6000 · corr pair ('AGEP', 'WKHP') · true corr 0.072 · TRTR F1 0.725

TSTR F1 across ϵ (higher = better; taller is more usable data)



independent pairwise aim mst

Correlation error (lower better)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.0521	0.0531	0.0530	0.0536	0.0535
pairwise	0.0543	0.0727	0.0468	0.0315	0.0202
aim ours	0.0808	0.0399	0.0609	0.0588	0.0621
mst ours	0.0326	0.0537	0.0413	0.0375	0.0522

TSTR F1 (higher better)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4397	0.4411	0.4305	0.4416	0.4616
pairwise	0.4664	0.4675	0.4894	0.4812	0.5219
aim ours	0.6493	0.6693	0.6910	0.6311	0.5668
mst ours	0.6487	0.6960	0.6984	0.6485	0.6069

Proved ϵ (context, not a contest)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4559	0.9122	1.8275	3.6639	7.3564
pairwise	0.4559	0.9122	1.8275	3.6639	7.3564
aim ours	0.3853	0.7782	1.5763	3.2011	6.5427
mst ours	0.3740	0.7518	1.5152	3.0681	6.2566

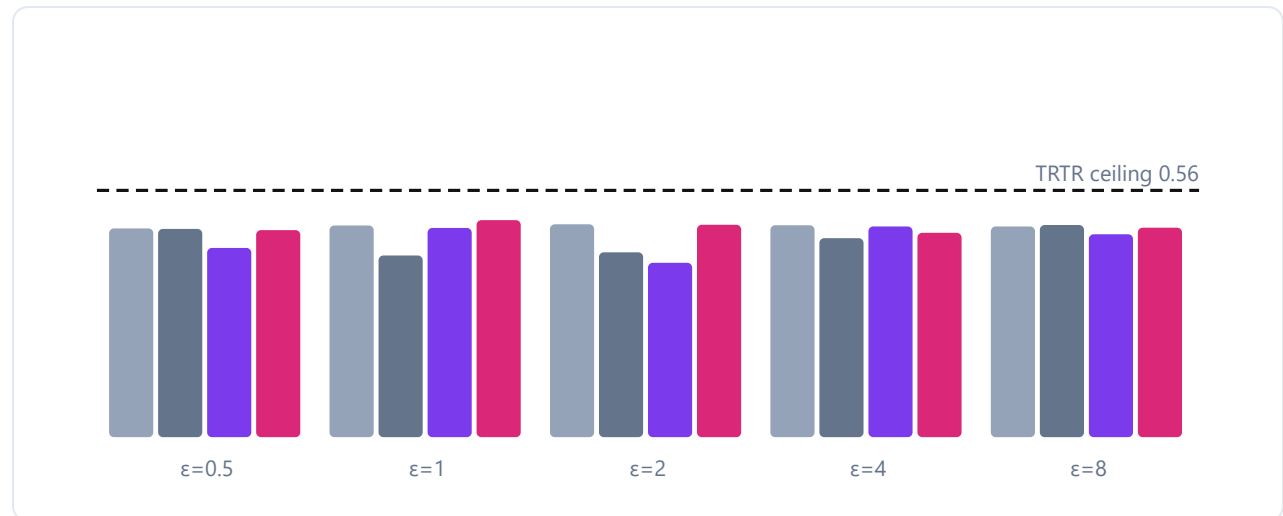
Membership AUC (all ≈ 0.5 = safe)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4987	0.5000	0.5028	0.5019	0.5032
pairwise	0.4941	0.4980	0.4896	0.4969	0.4971
aim ours	0.4916	0.4895	0.4795	0.4980	0.5064
mst ours	0.4861	0.4990	0.4858	0.4957	0.4958

UCI Bank Mktg

n=6000 · corr pair ('age', 'balance') · true corr 0.060 · TRTR F1 0.561

TSTR F1 across ϵ (higher = better; taller is more usable data)



independent pairwise aim mst

Correlation error (lower better)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.0450	0.0445	0.0446	0.0430	0.0416
pairwise	0.1485	0.0996	0.0968	0.0732	0.0289
aim ours	0.0517	0.0604	0.0516	0.0535	0.0579
mst ours	0.0552	0.0625	0.0538	0.0627	0.0576

TSTR F1 (higher better)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4747	0.4812	0.4841	0.4818	0.4792
pairwise	0.4733	0.4133	0.4205	0.4527	0.4824
aim ours	0.4301	0.4758	0.3968	0.4794	0.4617
mst ours	0.4706	0.4933	0.4829	0.4645	0.4763

Proved ϵ (context, not a contest)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4555	0.9115	1.8252	3.6578	7.3407
pairwise	0.4555	0.9115	1.8252	3.6578	7.3407
aim ours	0.3849	0.7772	1.5734	3.1944	6.5259
mst ours	0.3718	0.7460	1.5016	3.0313	6.1615

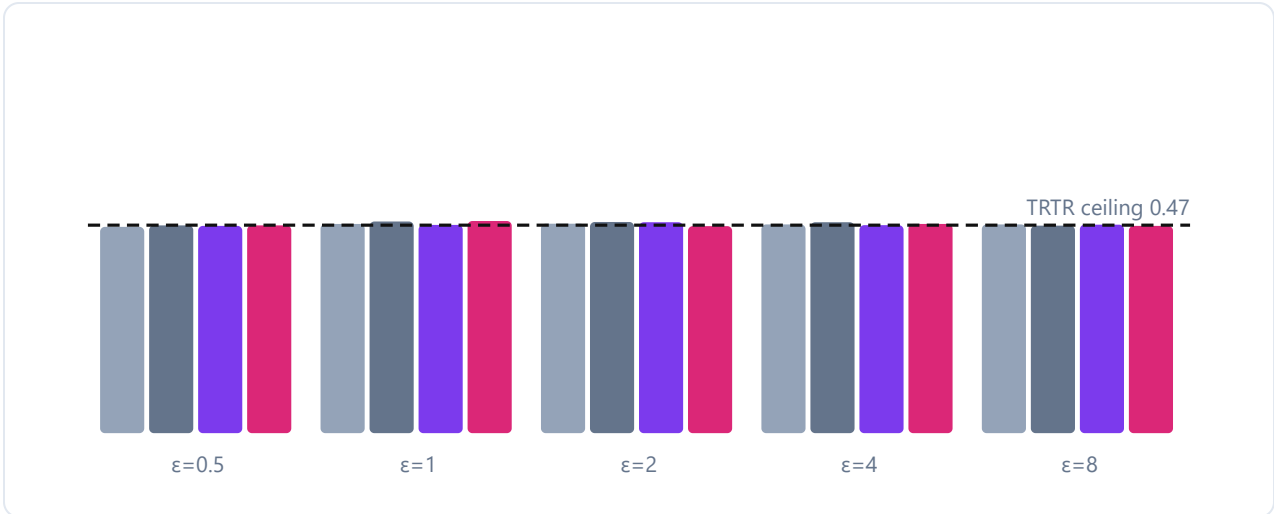
Membership AUC (all ≈ 0.5 = safe)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4983	0.4947	0.4926	0.4931	0.4943
pairwise	0.4836	0.4911	0.4959	0.4937	0.4986
aim ours	0.4929	0.4904	0.4843	0.4965	0.4884
mst ours	0.4840	0.4981	0.4994	0.4840	0.4910

UCI Diabetes 130 · healthcare

n=6000 · corr pair ('time_in_hospital', 'num_medications') · true corr 0.459 · TRTR F1 0.473

TSTR F1 across ϵ (higher = better; taller is more usable data)



independent pairwise aim mst

Correlation error (lower better)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4526	0.4529	0.4520	0.4525	0.4513
pairwise	0.3506	0.3551	0.3477	0.3572	0.3838
aim ours	0.1546	0.1096	0.1256	0.1176	0.1198
mst ours	0.1052	0.1125	0.1088	0.1108	0.1111

TSTR F1 (higher better)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4690	0.4760	0.4763	0.4748	0.4745
pairwise	0.4725	0.4815	0.4802	0.4797	0.4721
aim ours	0.4712	0.4734	0.4795	0.4732	0.4743
mst ours	0.4724	0.4824	0.4710	0.4760	0.4714

Proved ϵ (context, not a contest)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4572	0.9156	1.8357	3.6831	7.4021
pairwise	0.4572	0.9156	1.8357	3.6831	7.4021
aim ours	0.3869	0.7820	1.5857	3.2223	6.5923
mst ours	0.3771	0.7594	1.5343	3.1128	6.3602

Membership AUC (all ≈ 0.5 = safe)

mechanism	$\epsilon=0.5$	$\epsilon=1$	$\epsilon=2$	$\epsilon=4$	$\epsilon=8$
independent	0.4887	0.4928	0.4927	0.4898	0.4921
pairwise	0.5072	0.4965	0.5026	0.5101	0.5031
aim ours	0.5060	0.5031	0.5158	0.4978	0.5038
mst ours	0.5009	0.5037	0.5046	0.5004	0.4988

Source: results/*/h1_all_families.json + h1_mst*.json. Regenerate with scripts/gen_comparison_report.py. Literature ordering (AIM best on structure) reproduced, not the absolute numbers (our n=6000 subsample differs from the papers' full tables).